DEPLOYMENT & MODEL LOADING

Continuum OS: Software Deployment & LLM Loading

How the built Continuum OS software actually reaches Raspberry Pi 5 hardware, and the real, tested mechanism by which model weights are demand-paged into a running cluster.

Doc	ARCH-004
Status	Grounded in the built codebase at ~/continuum-os
Date	2026-09-19
Companion to	ARCH-001 (architecture), ARCH-002 (hardware/build), ARCH-003 (runtime diagram)

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Contents

00 Scope — what this document describes

01 From source tree to a bootable node image

02 Node provisioning — from generic image to a running role

03 continuumd's own boot sequence (PID 1)

04 The weight file — how a model is packaged

05 Loading a model into the cluster, step by step

06 Tiered page resolution and what's simulated vs. real

07 Honest status — what's verified, what isn't

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00 Scope — what this document describes

ARCH-002 specified a hardware/build plan; this document describes what actually happened when that plan was built. Every path, port number, struct field, and function name below is copied from the real source tree at ~/continuum-os, not re-derived from the architecture docs — where the build simplified, skipped, or simulated something the architecture called for, that gap is called out explicitly in §07 rather than papered over.

> **TWO QUESTIONS, TWO HALVES**
> §§01–03 answer "how does the software get onto the hardware" — from a monorepo checkout to a running continuumd supervising services on a Pi 5. §§04–06 answer "how does a model get into the cluster" — from a weight file on disk to resident, dequantized tensors inside a running shard-execd process, via swapd's demand-paging mechanism.

01 From source tree to a bootable node image

Continuum ships as one Buildroot-based node image, assembled in two stages: this repo's own CMake build compiles every daemon, then Buildroot's post-build.sh hook stages those binaries into the target root filesystem. The image is role-agnostic — the same image boots a control-plane node or a compute node; role is decided later, in §02.

01 Cross-compile every daemon first: cmake --preset aarch64-musl -S . -B build-aarch64 then cmake --build build-aarch64 --target continuumd node-agentd swapd shard-execd schedulerd api-gatewayd page-directoryd membershipd consoled

02 Point Buildroot at that output: CONTINUUM_BUILD_DIR=$(pwd)/build-aarch64 make, using image/buildroot/continuum_defconfig

03 Buildroot builds the kernel (with image/kernel/continuum-fragment.config's io_uring/userfaultfd/cgroup-v2 requirements) and the base rootfs, then invokes post-build.sh $TARGET_DIR

04 post-build.sh copies build-aarch64/kernel/continuumd to ${TARGET_DIR}/init — not a file under /opt, but literally /init, because BR2_INIT_NONE=y in the defconfig hands PID 1 straight to it

05 post-build.sh copies each of the eight service binaries into ${TARGET_DIR}/opt/continuum/bin/, and creates empty /run/continuum and /var/lib/continuum

06 post-build.sh copies the config *templates* (image/buildroot/config-templates/) into ${TARGET_DIR}/etc/continuum/ — deliberately the generic templates, not a filled-in config, so one image build is never accidentally specific to one node

07 Buildroot packages the result into a single flashable image

Fig. 1 — image/buildroot/post-build.sh's actual staging sequence (services/, kernel/src/main.c, image/buildroot/post-build.sh).*

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> **WHY CONTINUUMD IS `/INIT`, NOT A SYSTEMD UNIT**
>
> BR2_INIT_NONE=y in continuum_defconfig tells Buildroot not to install any init system at all. The kernel's own init= boot argument points at /init directly, and that file is continuumd itself. There is no systemd, no shell-driven init, and no package manager anywhere in the image — kernel/src/main.c's own header comment calls this out: continuumd "is PID 1 on every node", the thing that makes the distributed-OS framing literal rather than aspirational at this hardware tier.

The resulting rootfs layout, as staged by post-build.sh:

```
/init                    continuumd (PID 1)
/opt/continuum/bin/
  node-agentd  swapd  shard-execd  schedulerd
  api-gatewayd  page-directoryd  membershipd  consoled
/etc/continuum/          config templates (role-specific set copied in at §02)
/run/continuum/          created empty; sockets/pidfiles land here at runtime
/var/lib/continuum/      created empty; per-node persistent state
```

02 Node provisioning — from generic image to a running role

The image from §01 is identical for every board; a node's role — control-plane or compute — and its identity (node ID, peer addresses, ports, file paths) are filled in afterward, per image/buildroot/config-templates/README.md's own instructions: copy the config set matching a node's role into /etc/continuum/, then fill in node_id, peer IPs, and paths before first boot.

Which services a role runs

Each role's continuumd.conf names the exact children that role's continuumd supervises — read directly from the checked-in templates:

Role	continuumd supervises	Reference IP (6-node cluster)
control-plane (×3)	page-directoryd, membershipd, schedulerd, api-gatewayd, consoled	10.0.0.11 / .12 / .13
compute (×3)	swapd, node-agentd, shard-execd	10.0.0.21 / .22 / .23

Ports and sockets, as provisioned

Every value below is copied from the checked-in *.conf templates, not invented for this document:

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Service	Listen	Purpose
api-gatewayd	TCP :8080	OpenAI-compatible HTTP surface
schedulerd	unix:/run/continuum/schedulerd.sock	receives dispatch requests from api-gatewayd; routes by model name, e.g. route toy-model 10.0.0.21:7301
page-directoryd	Raft :7101, client :7111	3-way Raft quorum across control-plane peers (peer 1/2/3 10.0.0.1{1,2,3} 7101)
membershipd	Raft :7102, client :7112	a *separate* Raft group from page-directoryd, on purpose — a directory snapshot/compaction pause must never risk quorum membership stability
consoled	HTTP :7500, telemetry ingest :7401	operator console + node telemetry sink
swapd	unix:/run/continuum/swapd.sock	demand-paging service; also opens a shared /dev/shm DRAM pool and page table (§05)
node-agentd	TCP :7201	remote page fetch — how one compute node answers another's cache miss
shard-execd	TCP :7301	prompt dispatch / token streaming — deliberately TCP, not a Unix socket, since this is the one hop that crosses physical boards (see §07, bug #5)

CONFIG IS NEVER BAKED INTO THE IMAGE

post-build.sh's own comment is explicit about this: node-specific config "is NOT baked into the image — it's provisioned per node at flash/first-boot time." That's what keeps one Buildroot output usable for all six boards in the reference cluster, and what makes re-provisioning a replacement board a config copy, not a rebuild.

03 continuumd's own boot sequence (PID 1)

Once a node's kernel hands off to /init, kernel/src/main.c runs the same sequence on every node regardless of role — only the config file's service list differs.

01 Parse --config /etc/continuum/continuumd.conf (or -v for debug logging); exit if it fails to load

02 Open the boot-processor UART heartbeat link (cfg.uart_device) at 115200 8N1 — if the device isn't present, log a warning and continue with no hardware heartbeat rather than failing, so the same binary runs on a dev machine with nothing plugged in

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03 Block SIGCHLD/SIGTERM/SIGINT and pick them up via a signalfd, rather than a signal handler

04 Arm two timerfds: a 250ms supervision tick, and (only if the UART opened) a heartbeat_interval_ms-period timer, default 2000ms

05 supervisor_init() + supervisor_start_all() — fork/exec every service named in this node's config as a separate, fault-isolated child process

06 Enter the epoll loop: on SIGCHLD, reap and let the supervisor decide whether to restart; on the 250ms tick, run supervision bookkeeping; on the heartbeat timer, write one 'H' byte to the boot-processor UART

07 On SIGTERM/SIGINT: signal every child, wait up to 3s for a clean exit, then return

Fig. 2 — continuumd's actual startup, from kernel/src/main.c.

continuumd's own module comment is candid about one simplification worth repeating here: this reference build runs continuumd as an ordinary process rather than literally as kernel-assigned PID 1, so it can be developed and tested on an ordinary Ubuntu machine. The supervision, cgroup, and heartbeat logic is identical either way — only "mount /proc myself because I am PID 1" is skipped outside the real Buildroot image.

04 The weight file — how a model is packaged

ARCH-002 named a safetensors-style format as the eventual target. The actual build uses a much smaller custom format instead — lib/toymodel/include/toymodel_format.h's own comment explains why: a full safetensors reader is a lot of parsing machinery in service of proving one specific mechanism (demand-paging tensor bytes through swapd's tiered resolution), and that mechanism doesn't care what the header parser looks like — only that every tensor lands on a page boundary, which this format guarantees too.

Layout

A toymodel_header_t (magic CTMDLv1, page size, per-tensor scale, and one {start_page, n_pages} location per tensor) occupies page 0 in its entirety — read directly off disk by whoever loads it, since it's metadata, not part of the demand-paged region. Every tensor after it starts on its own page boundary, and each tensor's start_page is the *same* absolute page index that swapd's paging engine uses for its pread()s — no translation table between "which page holds tensor X" and "fetch page N" is needed anywhere in the system.

tools/gen_toy_weights.c writes this file for the model actually used in end-to-end testing — byte-level vocabulary (256 tokens, so no separate tokenizer), one attention block, one FFN block, int8 weights with a single shared dequantization scale of 0.05. Running it prints the exact page layout it wrote:

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Tensor	Shape	Bytes	Pages	Page range
header	—	4,096 (page 0, metadata only)	1	0
embedding	[256, 32]	8,192	2	1–2
w_q	[32, 32]	1,024	1	3
w_k	[32, 32]	1,024	1	4
w_v	[32, 32]	1,024	1	5
w_o	[32, 32]	1,024	1	6
w_ff1	[32, 64]	2,048	1	7
w_ff2	[64, 32]	2,048	1	8
unembedding	[32, 256]	8,192	2	9–10

Total: 11 pages × 4,096-byte page size = 45,056 bytes (44 KB) at 4-byte-per-dim/32-dim/64-ff_dim — trivially small on purpose, since proving the paging mechanism needs a model that's real, not one that's large. gen_toy_weights prints the exact tensor_region_bytes value a node's swapd.conf must be given to match, so the two never drift apart by hand-editing.

05 Loading a model into the cluster, step by step

This is the actual, tested path — gen_toy_weights → a compute node's swapd → shard-execd — as implemented in services/swapd/src/paging_engine.c, lib/pageclient/src/pageclient.c, and services/shard-execd/src/toy_model.c.

01 An operator (or provisioning script) runs gen_toy_weights <path> [seed] on the compute node, producing a page-aligned weight file at the path swapd.conf's weight_file will name

02 That node's swapd.conf sets weight_file, tensor_region_bytes (matching gen_toy_weights' printed total), page_size 4096, and this node's node_id

03 swapd starts under continuumd, opens the weight file read-only, and mmap()s an anonymous region of tensor_region_bytes — this region, not the file itself, is what gets demand-paged

04 swapd registers that region with userfaultfd() in MISSING mode (try_setup_real_uffd); it also creates the node's shared page table and DRAM pool under /continuum-pt-<node_id> and /continuum-pool-<node_id> (POSIX shared memory, lib/pagetable)

05 shard-execd starts, and pageclient_connect() attaches to that same shared page table and pool — retrying for a few seconds if swapd hasn't created them yet

06 toy_model_load() reads the header directly from the file (page 0 — metadata, never demand-paged) to learn every tensor's {start_page, n_pages}

07 For every page of every tensor, shard-execd calls pageclient_get_page(pc, page_index). A hit in the

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shared page table returns a pointer straight into the DRAM pool — zero-copy, no IPC

08 On a miss, pageclient sends a WIRE_PAGE_FETCH_REQ to swapd over /run/continuum/swapd.sock — the identical message a *remote* node's swapd would send on a cross-node miss

09 swapd's paging_engine_ensure_resident() touches one byte of that page inside the uffd-registered region; the kernel blocks the access and delivers a fault event to swapd's fault thread

10 The fault thread resolves the page's content: pread() from the local weight file at page_index * page_size first; only on a short/failed local read does it try each configured remote peer's swapd over TCP with the same WIRE_PAGE_FETCH_REQ/RESP pair

11 swapd answers the kernel fault with UFFDIO_COPY, copies the page into this node's shared DRAM pool, and records it in the shared page table at WIRE_TIER_LOCAL_DRAM with a last-touched timestamp for later LRU eviction

12 shard-execd's pageclient_get_page() call returns; toy_model_load() copies the raw int8 bytes out, multiplies every element by the header's shared scale (0.05), and moves to the next tensor

13 Once all eight tensors are loaded, toy_model_load() logs the total page count fetched and returns a ready toy_model_t — shard-execd can now run toy_model_forward_step() entirely against already-resident float arrays in its own process memory

Fig. 3 — one model load, from weight file on disk to resident float tensors inside shard-execd.

> **PLACEMENT IS A STATIC ROUTE, NOT AUTOMATIC SCHEDULING**
>
> Which compute node loads which model is decided by a line in that node's control-plane peer's schedulerd.conf — e.g. route toy-model 10.0.0.21:7301 — not by any automatic placement logic. api-gatewayd forwards a request to schedulerd over its Unix socket; schedulerd looks the model name up in that static route table and dispatches WIRE_DISPATCH_PROMPT to the matching shard-execd over TCP. Multi-shard pipeline placement across several compute nodes is the architecture's stated direction but isn't what this build's route table expresses today — it's a single-hop route per model name.

06 Tiered page resolution and what's simulated vs. real

lib/wire/include/wire.h defines four resolution tiers — WIRE_TIER_LOCAL_DRAM (1), WIRE_TIER_REMOTE_DRAM (2), WIRE_TIER_LOCAL_NVME (3), WIRE_TIER_REMOTE_NVME (4) — matching ARCH-001's tiering model. The actual resolution code in paging_engine.c is more direct than four separate code paths: resolve_page_content() tries the local weight file with one pread(), and on failure tries each configured remote peer with one uniform PAGE_FETCH_REQ/RESP exchange. The module's own comment explains the collapse: from the requester's side, "remote DRAM" and "remote NVMe" both just mean "ask a specific peer for this page" — the tier distinction matters for *how a peer answers*, not for how a page is requested, so this reference build doesn't duplicate that logic four ways.

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Tier	What ARCH-001 specifies	What this build does
Local DRAM	Page already resident in this node's DRAM pool	Exactly this — shm_pagetable_get() hit, zero-copy read (real)
Remote DRAM / Remote NVMe	Ask a peer node; peer answers from its own DRAM or NVMe depending on its own tier state	Folded into one path: WIRE_PAGE_FETCH_REQ to each configured peer in order, first usable WIRE_PAGE_FETCH_RESP wins (real, but tier-agnostic from the requester's side)
Local NVMe	Async read via io_uring	A synchronous pread() against the local weight file — the demand-paging *mechanism* (userfaultfd) is real; io_uring is named in ARCH-002 but not implemented, a swappable I/O-backend choice that doesn't touch the fault-handling path
userfaultfd itself	Real kernel demand paging	Real when available (UFFDIO_REGISTER succeeds) — falls back to an explicit page-touch-and-copy simulation (paging_engine_ensure_resident's non-uffd branch) when userfaultfd() is unprivileged or unavailable, so the same binary still runs correctly on a dev machine without CAP_SYS_PTRACE-adjacent permissions

07 Honest status — what's verified, what isn't

Carried over directly from the project's own README, since a deployment document that omits it would be misleading:

- The full CMake build compiles cleanly and 20 ctest tests pass, including multi-process end-to-end tests for paging, Raft, and console telemetry — on native x86_64/glibc. Cross-compiling to aarch64-linux-musl (the actual Pi 5 target) is written into the CMake presets but has not been exercised in this environment.
- The Buildroot image itself has never been run — no Buildroot checkout was attempted (a full build is hours and gigabytes, unlike the firmware below). image/ holds real, complete config: defconfig, kernel fragment, post-build script, per-role templates — just never exercised end to end.
- The boot-processor firmware, by contrast, was actually built against a real downloaded ARM toolchain and Pico SDK: zero compiler warnings, a byte-correct .uf2 (verified UF2 magic numbers, ELF32/ARM header, entry point in the RP2040's XIP

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region). What's unverified is that firmware running on real Pico hardware — none was available to flash.

- io_uring is a synchronous pread() for now (§06) — a named, deliberate simplification, not an oversight.
- No durable Raft persistence — lib/raftnet keeps its log purely in-memory, correct for consensus but a full quorum restart loses history.
- The KV cache lives in shard-execd's own process memory, not routed through swapd's paged tensor region — lib/kvcache's prefix-trie is wired in for prefix dedup, but not for the tier-aware KV swapping ARCH-001 describes.

> **WHY THIS LEVEL OF CANDOR**
>
> Every gap above was found by actually building and running the system — not by inspection. The Raft split-vote livelock, the blocking-timerfd hang, the jsmn double-count bug, and the Unix-socket-instead-of-TCP placement bug (this document's §02 already reflects the fix) were each caught by a specific end-to-end test failing. That's the argument for treating this document, and the diagram it accompanies, as a description of a system that has actually run — not a restatement of the plan that preceded it.

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